

Stochastic $M/M/c$ Erlang C Formulation & Analytic Tail Wait-Time Inversion Pipeline

Continuous Markov Chain Birth-Death Modeling $\rightarrow$ Erlang C Delay Probability $\rightarrow$ Invariant Order $t_{0.50} \leq t_{0.90}$

1. Parameter Ingestion

Station Hardware & Traffic

Operational Rates

Arrival rate: λ (veh/h)
Service rate: μ (veh/h/port)
Mean duration: $1/\mu$ (min)

Station Topology

Port capacity: $c \in \mathbb{N}^+$
Offered load: $a = \lambda/\mu$
Intensity: $\rho = \frac{\lambda}{c\mu} < 1$

Stability Condition

$\rho < 1.0 \Rightarrow$ Ergodic
Prevents infinite queueing

Load a, ρ

2. Erlang C Congestion

Analytical State Solution

Queue Probability

$P(\text{Queue}) = C(c, a)$
$$C(c, a) = \frac{\frac{a^c}{c!(1-\rho)}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!(1-\rho)}}$$

Exact closed-form expression

Waiting Time CDF

$W_q \sim$ Queue Wait Time
 $P(W_q \leq t) = 1 - C(c, a)e^{-c\mu(1-\rho)t}$
 $t \geq 0$ (Point mass at $t=0$)
 $P(W_q = 0) = 1 - C(c, a)$

CDF $P(W_q \leq t)$

3. Percentile Inversion

Analytic Tail Calculation

Tail Percentile t_p

$1 - C \cdot e^{-c\mu(1-\rho)t_p} = p$
$$t_p = \max\left(0, -\frac{\ln\left(\frac{1-p}{C(c, a)}\right)}{c\mu(1-\rho)}\right)$$

Eliminates simulation overhead

Operational Targets

Median: $p_{50} = \max\left(0, \frac{\ln(2C)}{c\mu(1-\rho)}\right)$
Tail: $p_{90} = \max\left(0, \frac{\ln(10C)}{c\mu(1-\rho)}\right)$
Tail wait time drives driver churn
Captures peak congestion delays

p_{50}, p_{90}

4. Invariant Ordering

Monotonicity & Delivery

Theorem 1 Invariant

$\forall \rho < 1 : t_{0.50} \leq t_{0.90}$
Case 1: $C \leq 0.10 \Rightarrow 0 \leq 0$
Case 2: $0.10 < C \leq 0.50 \Rightarrow 0 < t_{90}$
Case 3: $C > 0.50 \Rightarrow \ln 2C < \ln 10C$
Proven unconditionally

Downstream Delivery

Provenance Tag: ESTIMATED
728 Regimes Verified
0 Invariant Violations
Feeds ExplanationAgent
DES Ground Truth MAE: 9.39 min